# Six math clues · answer key

Transistors

Add the same number once for each group or row. Welcome a drawing or a spoken explanation.

### Math 1 · 9

3 + 3 + 3 = 9. Three transistors with 3 legs each make 9 legs.

### Math 2 · 20

5 + 5 + 5 + 5 = 20. Four rows of 5 make 20 tubes.

### Math 3 · 8

4 + 4 = 8. Two turns of 4 cups make 8 cups.

### Math 4 · 12

4 + 4 + 4 = 12. Three rows of 4 make 12 chips.

### Math 5 · 16

4 + 4 + 4 + 4 = 16. Four rows of 4 make 16 cards.

### Math 6 · 15

3 + 3 + 3 + 3 + 3 = 15. Five stencils of 3 shapes make 15 shapes.

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Six equal-group and array tasks use repeated addition. Rows and columns are each 2 to 5. Welcome drawings, spoken reasoning, or arranging small objects.

# Reading & writing · answer key

Transistors

### Reading · 1

A black bead with three legs: a tiny transistor.

### Reading · 2

Tubes got hot and burned out; transistors are small and cool.

### Reading · 3

A small signal on the middle leg opens a path for a bigger flow.

### Reading · 4

Other atoms let silicon carry electricity only when told.

### Reading · 5

On is one, off is zero; eight switches make 256 patterns.

### Reading · 6

Light prints them through stencils, layer after layer.

### Math · all

9, 20, 8, 12, 16, 15.

### Screen-free

Tubes came first; the middle leg opens the path; silicon is in sand; on and off are one and zero.

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Look, don't open: never take apart phones, chargers or batteries. The tap step needs a grown-up at the sink. Loose transistors are small; keep them from mouths.

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Offer dictation: let the child say a complete thought while you write it, then invite a picture or label.

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# Four looks at a switch

Transistors

### Quiz 1. What came before transistors?

Machines used glass vacuum tubes that glowed, got hot and burned out.

### Quiz 2. What does the middle leg do?

A tiny signal on the middle leg opens or closes the path for a bigger flow.

### Quiz 3. Where is silicon found?

Silicon is found in sand. A sprinkle of other atoms makes it a semiconductor.

### Quiz 4. What do on and off mean?

A computer treats on as one and off as zero.

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